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SUPPLEMENTARY MATERIAL

The Two Roots of the Tetrad Equation

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Abstract
The tetrad equation $g = e^T \eta\, e$ is quadratic in the frame, and like $x^2 = a$ it has a second root: if e solves it, so does $-e$. We prove that its solution set is the orbit of one frame under the Lorentz group, that an orientation sign splits it into exactly two classes, and that $-e$ lies in the class opposite to e if and only if the number D of spatial dimensions is odd. Because $\pm I$ is the centre of the Lorentz group, $-e$ is the only partner of e that every observer agrees on; spatial parity is how one observer sees it. Every quantity built from the metric takes one value on both roots, so the standard restriction to proper orthochronous frames, which keeps one root, is a choice that no metric equation makes. For $D = 3$ the two roots carry the two tetrahedra of the stellated octahedron, and the second is not a rotation of the first.

Keywords: tetrad, Lorentz group, root-blindness, spatial parity, stellated octahedron, proof

Setting

At a point of a spacetime with $1 + D$ dimensions and metric g of signature $(-, +, \dots, +)$, a *tetrad* is a real square matrix $e = (e^a{}_\mu)$ with

$$e^T \eta\, e = g, \qquad \eta = \mathrm{diag}(-1, I_D). \tag{1}$$

Equation (1) is quadratic in e, as $x^2 = a$ is in x. It defines an affine scheme over $\mathbb{R}$ [1, 2]; we study its real points, the solution set $S(g)$. The Lorentz group is $O(1, D) = \{\Lambda : \Lambda^T \eta \Lambda = \eta\}$, acting by $e \mapsto \Lambda e$. We write $P = \mathrm{diag}(1, -I_D)$ (spatial parity), $T = \mathrm{diag}(-1, I_D)$ (time reversal), $R = \mathrm{diag}(1, -1, 1, \dots, 1)$ (one spatial reflection), and

$$s(\Lambda) = \det \Lambda \cdot \mathrm{sgn}\, \Lambda^0{}_0. \tag{2}$$

$O(1, D)$ has four connected components, $X \cdot SO^+(1, D)$ for $X \in \{I, T, R, RT\}$, where $SO^+(1, D)$, the proper orthochronous group, is connected (for $D = 3$, see [3]).

Two Lemmas

Lemma 1 (time cone). *If $\Lambda \in O(1, D)$ and $\eta(u, u) < 0$, then $\mathrm{sgn}(\Lambda u)^0 = \mathrm{sgn}\, \Lambda^0{}_0 \cdot \mathrm{sgn}\, u^0$. Hence $\Lambda \mapsto \mathrm{sgn}\, \Lambda^0{}_0$ and s are homomorphisms onto $\{\pm 1\}$.*

Proof. From $\Lambda^T \eta \Lambda = \eta$ follows $\Lambda \eta \Lambda^T = \eta$, whose $(0, 0)$ entry gives $|r|^2 = (\Lambda^0{}_0)^2 - 1$ for $r = (\Lambda^0{}_i)$. A timelike u has $|\vec{u}| < |u^0|$. In $(\Lambda u)^0 = \Lambda^0{}_0 u^0 + r \cdot \vec{u}$ the second term has modulus below $|\Lambda^0{}_0 u^0|$, so the sign is that of $\Lambda^0{}_0 u^0$. Applying this to the column $u = \Lambda'_{\cdot 0}$, with $\eta(u, u) = -1$, gives $\mathrm{sgn}(\Lambda\Lambda')^0{}_0 = \mathrm{sgn}\, \Lambda^0{}_0\, \mathrm{sgn}\, \Lambda'^0{}_0$; det is multiplicative. $s(T) = 1$ and $s(R) = -1$. □

Lemma 2 (centre). *For $D \geq 1$, the only $Z \in O(1, D)$ commuting with all of $O(1, D)$ are $\pm I$, and the only $Y \in O(D)$ commuting with all of $O(D)$ are $\pm I_D$.*

Proof. Write Z in blocks $(\alpha, w^T; v, Y)$. It commutes with $\mathrm{diag}(1, S)$ for every $S \in O(D)$. With $S = -I_D$ this gives $v = w = 0$. Commuting with every diagonal sign matrix makes Y diagonal, and with every permutation matrix makes it λI_D; this also settles $O(D)$. A boost in the 0–1 plane then forces $\alpha = \lambda$, and $Z^T \eta Z = \lambda^2 \eta$ gives $\lambda = \pm 1$. □

The Two Roots

Theorem 1 (the solution set). *Let g be non-degenerate and $e \in S(g)$. Then:*

(a) *$S(g) = \{\Lambda e : \Lambda \in O(1, D)\}$, and $\Lambda \mapsto \Lambda e$ is a bijection. In particular $-e \in S(g)$.*

(b) *Fix an orientation and a time orientation, and set $o(e) = \mathrm{sgn} \det e \cdot \mathrm{sgn}(e\, t)^0$ for a future-directed timelike t. Then $o(\Lambda e) = s(\Lambda)\, o(e)$, so o splits $S(g)$ into exactly two classes S_+ and S_-, each the union of two components. T preserves each class and R exchanges them.*

(c) *$-e$ and Pe lie in the class opposite to e if and only if D is odd.*

Proof. (a) Every solution is invertible, since $(\det e)^2 \det \eta = \det g \neq 0$. For $e' \in S(g)$ put $\Lambda = e' e^{-1}$: then $\Lambda^T \eta \Lambda = e^{-T} g\, e^{-1} = \eta$. Conversely $(\Lambda e)^T \eta (\Lambda e) = g$, and $\Lambda e = \Lambda' e$ forces $\Lambda = \Lambda'$. (b) $u = e\, t$ is timelike, as $\eta(u, u) = g(t, t) < 0$; with $\det(\Lambda e) = \det \Lambda \det e$, Lemma 1 gives the rule. By (a), $S(g)$ is a copy of $O(1, D)$ with four components, on which s takes the values $s(I) = s(T) = 1$, $s(R) = s(RT) = -1$. Other choices of orientation change o by an overall sign only. (c) $s(-I) = (-1)^{D+1}(-1) = (-1)^D$ and $s(P) = (-1)^D$. □

Theorem 2 (the canonical root). *For $Z \in O(1, D)$, the map $e \mapsto Ze$ commutes with every change of frame, $\Lambda(Ze) = Z(\Lambda e)$ for all Λ and e, if and only if $Z = \pm I$. So $-e$ is the only root that every observer pairs with e. Parity is not central, and $Pe = T(-e)$: it is $-e$ with the time leg restored, the second root as seen by the observer of e.*

Proof. As e is invertible, the condition is $\Lambda Z = Z\Lambda$ for all Λ; apply Lemma 2. A boost B has $PBP = B^{-1} \neq B$, so P is not central, and $PT = -I$. □

Theorem 3 (root-blindness and root selection). *Every quantity determined by g, including the Christoffel symbols, the curvature, the Einstein tensor and the Einstein–Hilbert action, takes one value on all of $S(g)$; the same holds for tetrad fields, as $-e(x)$ solves (1) wherever $e(x)$ does. Restricting frames to $SO^+(1, D)\, e$, the usual choice for oriented and time-oriented frames [4, 5], keeps one component and excludes $-e$. Which component is kept is not determined by g.*

Proof. By (a), every element of $S(g)$ gives the same g, and a function of g cannot separate them. $-I \notin SO^+(1, D)$ since $(-I)^0{}_0 = -1$. All four components give the same g, so g cannot select one. □

The Two Tetrahedra

Take $D = 3$. Let E_a be the frame vectors of e, with $g(E_a, E_b) = \eta_{ab}$, and W the observer's rest space spanned by E_1, E_2, E_3. The four directions of Letter 1 [6] are $n_a = \sum_i M_{ia} E_i$, $a = 1, \ldots, 4$, with

$$M = \begin{pmatrix} +1 & +1 & -1 & -1 \\ +1 & -1 & +1 & -1 \\ +1 & -1 & -1 & +1 \end{pmatrix}, \tag{3}$$

the regular tetrahedron forced by the tetrad [7].

Theorem 4. *In the basis E_i of W:*

(a) *$n(-e) = n(Pe) = -M$ and $n(Te) = M$. A frame of the same observer, $\Lambda = \mathrm{diag}(\epsilon, Q)$, gives $n = Q^T M$ with $\det Q^T = s(\Lambda)$.*

(b) *No $Q \in SO(3)$ has $QM = -M$: no rotation carries the frame of e to that of $-e$ leg by leg. For a regular simplex M_D in $\mathbb{R}^D$, such a rotation exists if and only if D is even.*

(c) *The region $\mathrm{TetA} = \{x \in W : \max_a n_a \cdot x \leq 1\}$ of e becomes $-\mathrm{TetA} = \{\max_a(-n_a \cdot x) \leq 1\}$ for $-e$: the all-flipped region TetB of Letter 1. The two form the stellated octahedron [7].*

Proof. (a) The frame vectors of Λe are $E\Lambda^{-1}$: for $-e$ all four change sign, for Te only E_0, and for $\mathrm{diag}(\epsilon, Q)$ the spatial ones become $E'_j = \sum_i Q_{ji} E_i$; and $s(\Lambda) = \det Q$. (b) $\mathrm{rank}\, M_D = D$, so $QM_D = -M_D$ forces $Q = -I_D$, of determinant $(-1)^D$. (c) $(-n_a) \cdot x = n_a \cdot (-x)$. □

Remark (shape and place). The regular tetrahedron is congruent to its mirror image: the 90° rotation about the third axis maps the vertex set of TetA onto that of TetB, but reorders the legs. The second root changes neither shape nor metric, only the order of the legs (Theorem 4(b)) and the region of W occupied.

Scope

Ingredient	Without it
$\det g \neq 0$	For $g = 0$, $D = 1$: both 0 and $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ solve, and $S(g)$ is not one orbit
D odd	$-e$ stays in the class of e; for $D = 2$, $-M_2$ is M_2 turned by π
Ordered legs	As point sets TetB is a rotation of TetA

Summary

The tetrad equation keeps both roots, e and $-e$; no equation in the metric can tell them apart, so keeping one is a convention. In three dimensions they are the two tetrahedra of the stellated octahedron, which Letter 1 reaches independently by half-spaces.

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